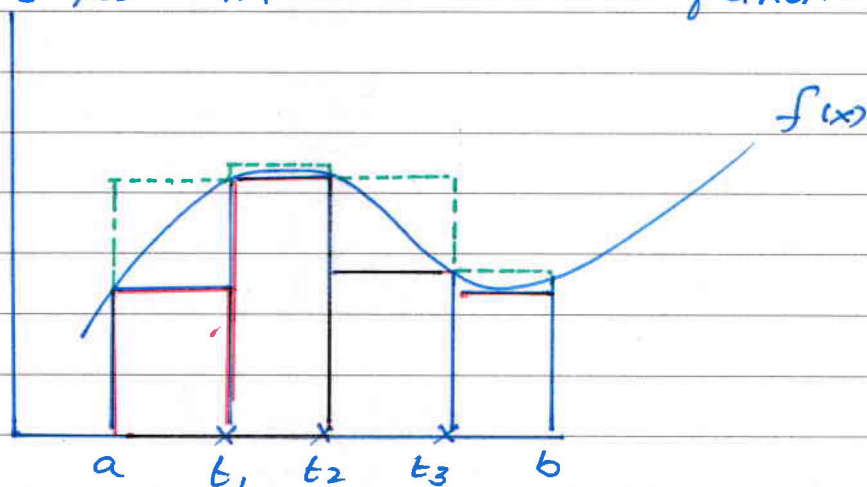


IV - Integration

We shall not develop this in detail
(See J.C. Burkill's book)

Let $I = [a, b]$ closed bounded interval
 $f: [a, b] \rightarrow \mathbb{R}$ is a bounded function



$\int_a^b f(x) dx$ is (intuitively speaking) the area between the graph of $f(x)$ the x axis bounded by the ordinates $x=a$ and $x=b$.

Problem is of course to define the notion of areas precisely.

Idea is simple: divide $[a, b]$ into smaller intervals $[a, t_1]$, $[t_1, t_2]$, $\dots$, $[t_{n-2}, t_{n-1}]$, $[t_{n-1}, b]$

Call $a = t_0$, $b = t_n$

$P: a = t_0 < t_1 < \dots < t_n = b$ is called a partition of $[a, b]$.

The inscribed rectangles (red) have

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total area

$$m_1(t_1 - t_0) + \dots + m_n(t_n - t_{n-1})$$

$$= \sum_{j=1}^n m_j(t_j - t_{j-1}) = L(P)$$

$$m_j = \text{g.l.b } \{ f(x) / t_{j-1} \leq x \leq t_j \}.$$

And.

The total area of the outer rectangles

$$(\text{green}) \text{ is } \sum_{j=1}^n M_j(t_j - t_{j-1}) = U(P)$$

$$M_j = \text{l.u.b } \{ f(x) / t_{j-1} \leq x \leq t_j \}.$$

Idea is to make the partition finer and finer adding more points in the subdivision (Partition)

and Calculating $L(P)$, $U(P)$

It is easy to Convince Yourself (Exercise)

Q is a partition obtained from P by adding one more point of subdivision:

additional point

$$Q: a = t_0 < \dots < t_k < s < t_{k+1} < \dots < t_n = b$$

(Do this with the picture on p 61 and see)

$$L(P) \leq L(Q) \leq U(Q) \leq U(P)$$

That is Upper approximations $U(P)$ diminish.

Lower approximations $L(P)$ increase

One can now interpolate more and more points...

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For Any two partitions:

A lower approx never crosses an upper approximation (why?). The two partitions may not be comparable.

Well, Assume P' , P'' are two different partitions not comparable to each other

Say $P' = \{0, \frac{1}{2}, 1\}$

for $[0, 1]$.

$$P'' = \{0, \frac{1}{3}, \frac{7}{5}, 1\}.$$

Look at $P' \cup P''$

$$L(P') \leq L(P' \cup P'') \leq U(P' \cup P'') \leq U(P'')$$

$$\therefore L(P') \leq U(P'')$$

$$L(P'') \leq U(P')$$

In some sense we should be looking at
 $\lim_{n \rightarrow \infty} L(P_n)$ or

$\sup \{ L(P) \mid P \text{ varies over all possible subdivisions} \},$

$\inf \{ U(P) \mid P \text{ varies over all possible subdivisions} \}.$

Let $P_1, P_2, P_3, \dots$ be a sequence of partitions obtained through the Method of bisection

$$P_1 = \{a, b\}$$

$$P_2 = \{a, \frac{a+b}{2}, b\} \text{ etc.}$$

$$P_3 = \left\{ a, \frac{1}{2} \left(a + \frac{a+b}{2} \right), \frac{a+b}{2}, \frac{1}{2} \left(\frac{a+b}{2} + b \right), b \right\}$$

...

We then have two sequences

$(L(P_n))$ monotone inc.

$(U(P_n))$ monotone dec.

$$L(P_n) \leq U(P_n) \quad \forall n$$

$\therefore$ Both Seq. are bounded and so $\lim_n L(P_n)$ and $\lim_n U(P_n)$ converge

Call the first one α

second one β

Evidently $\alpha \leq \beta$.

Def. If $\alpha = \beta$ then we say $f: [a, b] \rightarrow \mathbb{R}$ is (Riemann) integrable and the common value $\alpha (= \beta)$ is denoted by $\int_a^b f(x) dx$.

Thm 1: If $f: [a, b] \rightarrow \mathbb{R}$ is bounded and has finitely many discontinuities then $\int_a^b f(x) dx$ exists.

For example $f(x) = \begin{cases} \sin \frac{1}{x} & \text{if } 0 < x \leq 1 \\ 0 & \text{if } x = 0. \end{cases}$
 $\int_0^1 f(x) dx$ exists.

Thm 2: If $f: I \rightarrow \mathbb{R}$ is differentiable throughout I and f' is continuous

Then for $a, b \in I$ with $a < b$,

$$\int_a^b f'(x) dx = f(b) - f(a).$$

Def: If $f: I \rightarrow \mathbb{R}$ is Continuous and $a \in I$.

The function $F(t) = \int_a^t f(x) dx : t \in I, t > a$

is called a primitive of f

Convention: $\int_a^a f(x) dx = 0$.

if $b < a$ we declare $\int_a^b f(x) dx = -\int_b^a f(x) dx$

With this one can define the primitive of a Cont. function $f: I \rightarrow \mathbb{R}$ to be

$$F(t) = \int_a^t f(x) dx \text{ for all } t \in I$$

Thm3: If $f: I \rightarrow \mathbb{R}$ is Continuous

Then, its primitive $F(t) = \int_a^t f(x) dx$

is differentiable and $F'(t) = f(t)$.

Comments: In Thm 2 we are demanding

that f' is Continuous

for concluding $\int_a^b f'(t) dt = f(b) - f(a)$

Continuity of f' is an important in Thm 2.

In Thm 3: We are demanding f is Continuous to conclude

$$\frac{d}{dt} \int_a^t f(x) dx = f(t)$$

Proofs of these things will be done
Only in MA403 (real Analysis)
Cannot be done here. Require subtler ideas
on limits - See J.C. Burkill...

Exercise: Find the area under the
graph of $f(x) = x^3 + 3x - 1$
bounded by x axis between ordinates
 $x = 0$ and $x = 1$.

Solid

Volume of a ~~Solids~~ of revolution.

Suppose $f: [a, b] \rightarrow \mathbb{R}$ is positive
Find the volume of the solid obtained
by revolving the graph $y = f(x)$ about
the x -axis between ordinates
 $x = a$ and $x = b$.

Can this be obtained as a limit of a
monotone increasing seq L_n ?

Will this limit be $\pi \int_a^b (f(x))^2 dx$?

Discuss.

Find the volume of the sharp needle
obtained by revolving the graph $y = x^{10}$
between ordinates $x = 0$, $x = 1$.

Shell Method and Washer Method

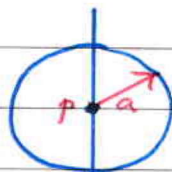
The method will be explained through
a couple of simple examples.

Find the volume of a bicycle tube with cross sectional radius a and mean longitudinal radius A ($A > a$)

The circle

$$x^2 + (y-A)^2 = a^2$$

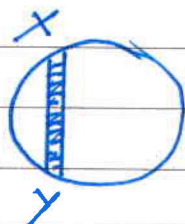
is rotated about the x -axis. The solid is the bicycle tube (known as the torus)



$$P = (0, A)$$

Vertical

Take a thin slice of the circle with ends X, Y



The vertical slice has x -coordinate x
then $X = A + \sqrt{a^2 - x^2}$
 $Y = A - \sqrt{a^2 - x^2}$

This thin vertical slice when rotated gives a washer with volume

$$= \pi \delta x \left((A + \sqrt{a^2 - x^2})^2 - (A - \sqrt{a^2 - x^2})^2 \right)$$

δx = thickness of the washer.

$$= \pi(\delta x) \cdot 4A \sqrt{a^2 - x^2}$$

Sum of the volumes of these washers =
Vol. of the torus.

The vertical slice XY moves from left end with x -coord $-a$ to right end w/ x -coord a

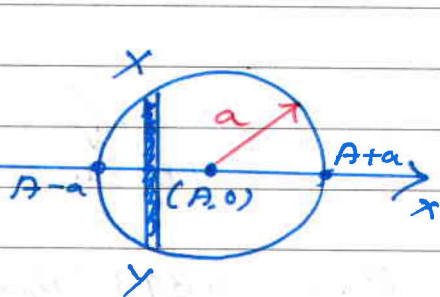
$$\begin{aligned} \text{Vol} &= \int_{-a}^a 4\pi A \sqrt{a^2 - x^2} dx \\ &= 8\pi A \int_0^a \sqrt{a^2 - x^2} dx = 2\pi^2 A a^2 \end{aligned}$$

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Now let us approach the same problem differently.

Think of the bicycle tube as being obtained by rotating $(x-A)^2 + y^2 = a^2$ about the y axis

Again we take a vertical slice XY of thickness δx with x-coordinate x
y-coord of X, Y are $\pm (a^2 - (A-x)^2)^{1/2}$



So that length $(XY) = 2 \sqrt{a^2 - (A-x)^2}$

This vertical slice XY generates a cylindrical shell with volume
 $= 2\pi x \delta x \cdot 2 (a^2 - (A-x)^2)^{1/2}$

These shells together form the region inside the bicycle tube.

Volume of the tube

$$= 4\pi \int_{A-a}^{A+a} x \sqrt{a^2 - (A-x)^2} dx$$

$A-a \qquad A-a < x < A+a$

put $x - A = a \sin \theta$

$$Vol = 4\pi \int_{-\pi/2}^{\pi/2} (A + a \sin \theta) a^2 \cos^2 \theta d\theta$$

$$= 8\pi A a^2 \int_0^{\pi/2} \cos^2 \theta d\theta = 2\pi^2 A a^2$$

Thomas Finney

Ex: Work out Ex: pages 382 - 393

Work out all the Solved and unsolved problems in Thomas-Finney on Shell / Washer Method.

pages 382-393

Contains several illuminating examples.

9th Ed.

Ex: p 393 Thomas Finney (9th Ed)

$$f(x) = \frac{\tan^2 x}{x} \quad \text{if } 0 \leq x \leq \pi/4$$

$$f(0) = 0.$$

Find the volume of the solid obtained by revolving the graph about x -axis.

Ex: p 393: Thomas Finney (9th Ed)

$$x = 3y^2 - 2$$

$$x = y^2$$

and

Region between these curves, x axis is revolved around the x -axis.

Find the volume of solid of revolution using Shell Method / Washer Method.

Do problem 38 on p 393 of Thomas Finney.
(9th Ed)

Additional problems: (Based on Chapters 1 to 4)

(1) prove that if $x_{n+1} - x_n \rightarrow 0$
then $\frac{x_n}{n} \rightarrow 0$.

(2) $S_n = \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \dots + \frac{1}{n(n+1)}$

Show that (S_n) Converges

Prove that if $t_n = 1 + \frac{1}{2^2} + \dots + \frac{1}{n^2}$

then (t_n) Converges.

(3) prove that

$$x_n = 1 + \frac{1}{2} + \dots + \frac{1}{n} - \log(n+1)$$

then (x_n) is monotone increasing

Hint: $e^x > 1+x$ if $x > 0 \dots$

(4) Show that (x_n) in the preceding exercise is bounded.

Hint: Apply LMVT to $\log x$ on $[k, k+1]$
($k = 1, 2, \dots, n$)

(5) The $\lim_n x_n$ is denoted by γ (Called the

Euler-Mascheroni Constant). It is not known whether γ is rational or irrational. For a very interesting historical ^{notes} ~~account~~ see G. Chrystal: Algebra, vol II. p 81

(5) Let $f(x) = \begin{cases} e^{-\frac{1}{2}x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0. \end{cases}$

prove that $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$.

(6) Start with

$$\sin^{2n+1} x < \sin^{2n} x < \sin^{2n+1} x \quad ; n \in \mathbb{N}.$$

Integrate from 0, to $\pi/2$.

Use Sandwich-Thm to deduce that -

$$\pi = 2 \lim_{n \rightarrow \infty} \frac{2 \cdot 2 \cdot 4 \cdot 4 \dots 2m \cdot 2m}{3 \cdot 3 \cdot 5 \cdot 5 \dots 2m-1 \cdot 2m+1}$$

This is Wallis's formula (Arithmetica Infinitorum).

See J.C. Burkill p134

(7) Compute $\lim_{n \rightarrow \infty} \left\{ \frac{(n+1) \dots (n+n)}{n!} \right\}^{1/n}$

(8) Compute $\lim_{x \rightarrow 0} \left(\frac{1}{2} (a^x + b^x) \right)^{\frac{1}{x}} ; (a, b > 0)$

First recall the def of p^2 when $p > 0$ and $q \in \mathbb{R}$.

Use the fact that the exponential function is Continuous. First Compute $\lim_{x \rightarrow 0} \frac{1}{x} \log \left(\frac{a^x + b^x}{2} \right)$

(9) A rectangle is inscribed in the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1. \quad \text{What is the Maximum possible area of such a$$

~~rectangle?~~ rectangle?

(10) Do Ex 10 on p 135 of J.C. Burkill.

prove $\int_0^1 x^m (x^2-1)^m x^n (x^2-1)^n = 0$ if $m \neq n$

72.1

(11) Let $f: I \rightarrow \mathbb{R}$ be a convex function

I is an open interval.

Can f have a local Max?

Can f have a local Min?

Can f have two distinct local Min?

(12) Let $f: [0, \infty) \rightarrow \mathbb{R}$.

We say $\lim_{x \rightarrow \infty} f(x) = l$ if given $\epsilon > 0$
(Error tolerance)

$\exists x_0 \in \mathbb{R}$ such that
 $f(x) \in (l - \epsilon, l + \epsilon)$ for all $x > x_0$.

Show that

$$\lim_{x \rightarrow \infty} e^{-x^2} = 0, \quad \lim_{x \rightarrow \infty} \frac{1}{1+x^2} = 0$$

What is $\lim_{x \rightarrow \infty} \tan^{-1} x$?

(13) Assume $f: [0, \infty) \rightarrow \mathbb{R}$ is

continuous. $\int_0^{\infty} f(t) dt$ is by def

$\lim_{x \rightarrow \infty} \int_0^x f(t) dt$ if the limit exists.

prove: If $f \geq 0$ and continuous

$\int_0^{\infty} f(t) dt$ exists iff $\int_0^x f(t) dt$

is bounded above for $x > 0$.

Does $\int_0^{\infty} e^{-t} t^{a-1} dt$ exist for $a \geq 1$?

Calculate its value when
 $a \in \mathbb{N}$.

72.2

(14) Suppose $f: [a, b] \rightarrow \mathbb{R}$ is continuous. Prove that there is a $c \in [a, b]$ such that

$$\int_a^b f(t) dt = f(c)(b-a)$$

Hint:

$$L_n \geq m(b-a)$$

$$U_n \leq M(b-a)$$

$$m = \inf f \text{ and } M = \sup f \text{ on } [a, b]$$

$$m(b-a) \leq \int_a^b f(t) dt \leq M(b-a)$$

$$\therefore m \leq \frac{1}{b-a} \int_a^b f(t) dt \leq M$$

Now use Theorems B, C of Chapt 2

Result of this exercise is called the first MVT of integral Calc.

There is a second one due to Ossian Bonnet that we shall not discuss.

(15) Use (14) to prove that if

$f: [a, b] \rightarrow \mathbb{R}$ is continuous then

$G(t) = \int_a^t f(x) dx$ is diff and

$$G'(t) = f(t).$$

$$\frac{G(p+h) - G(p)}{h} = \frac{1}{h} \int_p^{p+h} f(x) dx : f(p) = \frac{1}{h} \int_p^{p+h} f(x) dx$$

$$\frac{G(p+h) - G(p)}{h} - f(p) = \frac{1}{h} \int_p^{p+h} (f(x) - f(p)) dx$$

$$\left| \frac{G(p+h) - G(p)}{h} - f(p) \right| \leq \frac{1}{h} \int_p^{p+h} |f(x) - f(p)| dx$$

c lies between p and $p+h$

so $f(c) \rightarrow f(p)$ as $h \rightarrow 0$.

Tut 3

- 72.3 -

(16) prove that if $f(t) = \begin{cases} e^{-1/t^2} & t \neq 0 \\ 0 & t = 0 \end{cases}$

then f is diff. at 0.

In fact prove that f is ~~diff~~ k -times diff at 0 for every $k \in \mathbb{N}$.

(17) For a bounded function $f: I \rightarrow \mathbb{R}$

$$\sigma(I) = \{ \text{l.u.b. } \{ f(x) / x \in I \} - \text{g.l.b. } \{ f(x) / x \in I \} \}$$

If $J \subset I$ Compare $\sigma(J)$ and $\sigma(I)$

Suppose J_n is a seq of closed intervals s.t. $p \in J_n$ for all n ; length $J_n \rightarrow 0$

Is it true that $\sigma(J_n) \rightarrow 0$ as $n \rightarrow \infty$.

Try for example $f(x) = \begin{cases} \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$ ($p=0$)

Try it with $f(x) = \cos x$ on $(-\pi, \pi)$; ($p=0$)

Try it with $f(x) = \begin{cases} 1 & x > 0 \\ -1 & x \leq 0 \end{cases}$ ($p=0$)

Is it true that $\sigma(J_n) \rightarrow 0$ if f is continuous at p .
proof?

$\sigma(I)$ is called the spread of f on I

(More common terminology is oscillation of f at p but the concept has nothing to do with ~~oscill~~ any sinusoidal behaviour. So the word spread is sometimes used to avoid confusion. See for example

F. Bowman and F.A. Gerard: Higher Calc. Camb. Univ Press)

Tut 3

(18) Let I be an interval $[a, b]$; $a, b \in \mathbb{R}$

$$I_2 = I, \quad I_3 = \left[a, \frac{a+b}{2} \right], \quad I_4 = \left[\frac{a+b}{2}, b \right]$$

$$I_5 = \left[a, \frac{3a+b}{4} \right], \quad I_6 = \left[\frac{3a+b}{4}, \frac{a+b}{2} \right]; I_7 = \dots; I_8 = \dots$$

(Note the repeated bisection producing 2^{n-1} new intervals at each stage).

We have a seq of intervals $I_2, I_3, I_4, I_5, I_6, I_7, I_8, \dots$

If $f: I \rightarrow \mathbb{R}$ is continuous prove that

$$\sigma(I_n) \rightarrow 0 \text{ as } n \rightarrow \infty$$

Suppose not. Then $\exists c > 0$ such that

$$\sigma(I_n) \geq c \text{ for infinitely many indices.}$$

namely for infinitely many of the intervals.

(at least)

$$\text{One of the halves } [a, \frac{a+b}{2}] \text{ or } [\frac{a+b}{2}, b]$$

must contain infinitely many

$$\text{of these } I_n \text{ for which } \sigma(I_n) \geq c.$$

Pick that one and call it J_1

$$\text{Bisect } J_1 = J_1' \cup J_1''$$

(at least)

One of the halves J_1', J_1'' must contain infinitely many of the I_n s.t. $\sigma(I_n) \geq c$ pick that one and call it J_2

proceeding thus we have a nested seq

$$J_1 \supset J_2 \supset J_3 \supset \dots \text{ length } J_n = \frac{1}{2^n} (b-a)$$

So left endpoints $\rightarrow p$.

Right endpoints $\rightarrow p$. $\therefore p \in J_n$ for all $n \in \mathbb{N}$

$$\sigma(J_n) \rightarrow 0 \text{ as } n \rightarrow \infty \text{ due to Continuity of } f$$

But J_n contains infinitely many I_k s.t.

$$\sigma(I_k) \geq c$$

$$\therefore \sigma(J_n) \geq \sigma(I_k) \geq c$$

This is a contradiction since $\sigma(J_n) \rightarrow 0$.

~~(18) Thus we have proved that if $f: [a, b] \rightarrow \mathbb{R}$ is continuous~~

Tut 3

(18) Let (I_n) be the sequence in the last Ex.

$$A_1 = \sigma(I_2); \quad A_2 = \max(\sigma(I_3), \sigma(I_4))$$

$$A_3 = \max(\sigma(I_5), \sigma(I_6), \sigma(I_7), \sigma(I_8))$$

etc;

prove that $A_n \rightarrow 0$ as $n \rightarrow \infty$

This follows at once from (17).

72.5 (Optional Exercise)

(19) Thm E: Prove that a continuous function is Riemann integrable.

Thm F: If $f: [a, b] \rightarrow \mathbb{R}$ is such that f' exists and it is continuous on $[a, b]$ (namely f is actually defined on a slightly larger interval $[a-c, b+c]$ and is diff. on $(a-c, b+c)$).

$$\int_a^b f'(t) dt = f(b) - f(a).$$

Recalling the def of the integral

$$L_n \rightarrow L : U_n \rightarrow U : L \leq U$$

If f is not integrable then $L \neq U$.

$$\text{put } \alpha = U - L$$

$$L_n \leq L < U \leq U_n \Rightarrow U_n - L_n \geq U - L = \alpha > 0$$

$$0 \leq U_n - L_n = \frac{(b-a)}{2^n} \sum_{j=1}^{2^n} \sigma(I_j) \leq A_n (b-a) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

(by Ex 18)

This is a Contradiction since $U - L = \alpha > 0$.

proof of Thm F:

Let $L_n =$ Lower Riemann Sum for $\int_a^b f'(t) dt$
wrt dyadic partitioning

$$L_n = \sum f'(c_j) (t_j - t_{j-1}), \quad : f'(c_j) = \inf \text{ of } f' \text{ on } [t_{j-1}, t_j].$$

~~=~~ By LMVT

$$f(t_j) - f(t_{j-1}) = f'(c_j) (t_j - t_{j-1})$$

$$\therefore L_n = \left| \sum (f'(c_j) - f'(c_{j-1})) (t_j - t_{j-1}) \right| \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$+ \sum (f(t_j) - f(t_{j-1}))$$

$$\downarrow \int_a^b f'(t) dt$$

$$\therefore \int_a^b f'(t) dt = \underline{\underline{f(b) - f(a)}}$$